ultramicroscopic precision. This is why general relativity works on large enough distance (and time) scales—the scales relevant for many typical astronomical applications—but is rendered inconsistent on short distance (and time) scales. The central tenet of a smooth and gently curving geometry is justified in the large but breaks down due to quantum fluctuations when pushed to the small.

The basic principles of general relativity and quantum mechanics allow us to calculate the approximate distance scales below which one would have to shrink in order for the pernicious phenomenon of Figure 5.1 to become apparent. The smallness of Planck's constant—which governs the strength of quantum effects—and the intrinsic weakness of the gravitational force team up to yield a result called the Planck length, which is small almost beyond imagination: a millionth of a billionth of a billionth of a billionth of a centimeter (10-33 centimeter).7 The fifth level in Figure 5.1 thus schematically depicts the ultramicroscopic, sub-Planck length landscape of the universe. To get a sense of scale, if we were to magnify an atom to the size of the known universe, the Planck length would barely expand to the height of an average tree.

And so we see that the incompatability between general relativity and quantum mechanics becomes apparent only in a rather esoteric realm of the universe. For this reason you might well ask whether it's worth worrying about. In fact, the physics community does not speak with a unified voice when addressing this issue. There are those physicists who are willing to note the problem, but happily go about using quantum mechanics and general relativity for problems whose typical lengths far exceed the Planck length, as their research requires. There are other physicists, however, who are deeply unsettled by the fact that the two foundational pillars of physics as we know it are at their core fundamentally incompatible, regardless of the ultramicroscopic distances that must be probed to expose the problem. The incompatibility, they argue, points to an essential flaw in our understanding of the physical universe. This opinion rests on an unprovable but profoundly felt view that the universe, if understood at its deepest and most elementary level, can be described by a logically sound theory whose parts are harmoniously united. And surely, regardless of how central this incompatibility is to their own research,